

Amin Sabir

+447450 179667 | amintuf@hotmail.com |  Website

 LinkedIn |  GitHub |  Google Scholar

EDUCATION

- **PhD, University of Bath** 2024 - Present
Mathematics PhD in numerical analysis & optimisation for imaging inverse problems
Focus on developing data-driven regularisation techniques for mixed noise models using plug-and-play and infimal convolution for light microscopy & medical imaging
Bath, UK
- **MRes in Statistical Applied Mathematics, University of Bath** 2023 - 2024
Distinction, specialising in data-driven regularisation techniques for inverse problems
Averaged over 85% in Numerical Optimisation, Numerical Linear Algebra and Data Filtering & Assimilation
Bath, UK
- **MSci in Mathematics, University College of London (UCL)** 2018 - 2022
First Class Honours, Dissertation: Analysis of asymptotic series expansions for ODEs
94% Computational Methods & Programming, 88% in Probability & Statistics
London, UK

WORK EXPERIENCE

- **University of Bath and Industry Partners**  01/2024, 07/2024, 01/2025
Industry inter-disciplinary researcher at Integrative Think-Tanks (ITTs)
Bath, UK
 - Contributed in three week-long industry ITTs with Wessex Water, Rolls-Royce and Diamond Light Source
 - Tackled industry challenges ranging from modelling bacteria contamination and river flows to developing deep learning techniques for imaging microscopy techniques of material samples
 - Presented findings to a panel of over 50 academics and industry experts, resulting in a £200,000 multi-year research grant partnership
- **Infosys** 06/2021 - 09/2021
Internship - Software Developer
London, UK
 - Led a network data analysis project to identify network faults on a web server
 - Analysed network features data and their patterns using Python to build a functioning ML model (XGBoost) for fault classification
 - Applied hyperparameter tuning techniques to improve model accuracy from 71% to 97%
 - Presented my findings to an expert panel of seven senior professionals within the Networks and ML team
- **SparxMaths (HegartyMaths)** 05/2019 - 09/2019
Educational content developer
London, UK
 - Utilised Python and Latex to produce GCSE and A-level Mathematics content
 - Applied Autograph and Inkscape design software to create a diverse set of graphical questions
 - Identified problem reports, ran solution fixes and tested functionality for the website question bank

RESEARCH PROJECTS

- **Industry Inter-disciplinary Researcher - Rolls-Royce Holdings** 06/2024 - 07/2024
Project: Reverse engineering atmospheric dust from aircraft engine samples
Code  
 - Managed a think-tank project on the effects of atmospheric dust on aircraft engines
 - Developed mathematical models based on Bayesian statistics and tested on a real aircraft engine dataset
 - Published a GitHub repository with practical Python code and a technical report, bringing further ongoing collaboration between the University of Bath and Rolls-Royce
- **Undergraduate Research Assistant - UCL Division of Medicine** 08/2018 - 03/2020
Project: Effects of normobaric hypoxia on oxygen saturation variability
 - Undertook a research project on finding a suitable model to measure oxygen variability in haemoglobin
 - Implemented a numerical model in Python based on physiological processes
 - Applied the model to real medical data and contributed to a published academic paper on the effects of Hypoxia in the body for the High-Altitude Medicine & Biology Journal

PUBLICATIONS

[1] J.T. Costello, A.S. Bhogal, T.B. Williams, R. Bekoe, **A. Sabir**, M.J. Tipton, J.O. Corbett, A.R. Mani (2020). **Effects of Normobaric Hypoxia on Oxygen Saturation Variability**. *High Altitude Medicine & Biology*, Vol. 21, No. 1.

TALKS AND PRESENTATIONS

IPMS: Inverse Problems Modelling & Simulation conference <i>Contributed talk: "Learned regularisation for light-sheet fluorescence microscopy"</i>	June 2026 Mellieha, Malta
REMODEL: Research Exchanges in Mathematics of Deep Learning conference <i>Contributed talk: "Learned regularisation for light-sheet fluorescence microscopy"</i>	May 2026 Trondheim, Norway
Symposium on AI and Reconstruction for Biomedical Imaging <i>Poster: "Advanced regularisation methods for light-sheet fluorescence microscopy"</i>	March 2026 London, UK
LMS Conference on Medical Imaging & AI <i>Poster: "Learned infimal convolution regularisation for inverse problems"</i>	June 2025 Bath, UK

ORGANISATION OF SCIENTIFIC EVENTS

Co-lead Organiser — SIAM Postgraduate Conference <i>Society for Industrial and Applied Mathematics annual postgraduate conference</i>	November 2025 Bristol, UK
Lead Organiser — Statistical Applied Mathematics Conference <i>Two-day conference with 50+ participants at University of Bath</i>	July 2025 Bath, UK

TEACHING AND SUPERVISION

Module Teaching Assistant — tutorials and programming labs <i>Multivariate Calculus, Python programming modules</i>	2023/2024, 2024/2025, 2025/2026 University of Bath
Module Co-advisor and Instructor <i>Masters level course on inverse problems theory and its application</i>	2024/2025 University of Bath
Undergraduate Summer Project Supervision <i>Supervised an undergraduate student on "Combining classical and learned regularisation for inverse problems"</i>	Summer 2025 University of Bath

ADDITIONAL SKILLS

Coding: Python (PyTorch, NumPy, Pandas) **Tools:** Version control (Git), HPC (SLURM), LaTeX
Activities and volunteering: Long-distance running (3 marathons, 15 half-marathons), charity fundraising (raised over £12,000 for Epilepsy Society), athletics volunteering for disadvantaged children in Westminster